Haoyao Ruan, PhD Candidate

haoyao.ruan@uta.edu
linkedin.com/in/haoyao-ruan
github.com/haoyao-ruan

Available June 2027 (estimated) • TN-eligible (USMCA Statistician)

Summary

Biostatistician with 3.5+ years of full-time industry statistics (ERM, Cytel) and a PhD due summer 2027. At Cytel, built oncology evidence for pharma sponsors: indirect treatment comparisons (MAIC, STC), external and synthetic control arms with **Bayesian borrowing**, and survival analysis. Co-author of three oncology papers in *Advances in Therapy* and *J Comp Eff Res*. First author of a Bayesian methods paper accepted at ***JRSS Series A*** (2026). Expert in R, Shiny and MCMC; working SAS.

Experience

Summer 2026 — **Biostatistician III (Intern)**
Novartis, Advanced Quantitative Sciences • New Jersey, USA
- Built a **Shiny app** to track **survival-data maturity** and predict events in **oncology trials**.

Sep 2021 – Aug 2023 — **Research Consultant, Statistics (Full-time)**
Cytel Inc., Real-World Evidence and Advanced Analytics • Toronto, ON, Canada
- Ran **indirect treatment comparisons (MAIC, STC)** for oncology submissions, in relapsed/refractory multiple myeloma (*Adv Ther*, 2022) and BRCA1/2+ metastatic castration-resistant prostate cancer (*Adv Ther*, 2024).
- Applied **Bayesian borrowing** to a 1L NSCLC **external control arm** (*J Comp Eff Res*, 2024).
- First author on the variance–bias trade-off of covariate adjustment in **synthetic control methods** (*Value in Health*, 2022).
- Worked with principal statisticians to turn new methods literature into applied oncology analyses, including **survival models**.

Oct 2019 – May 2021 — **Statistical Consultant (Full-time)**
Environmental Resources Management (ERM) • Toronto, ON, Canada
- Built **multivariate models and statistical tests** for environmental and health risk assessments, including BACI designs, GAMMs and bootstrap kernel density estimates.
- Automated data pipelines and **Shiny/R Markdown dashboards**, cutting reporting time by **60%**. Received the **High Five Award** for statistical reporting.

2019 – 2023 — **Research Assistant (Part-time)**
Simon Fraser University, Department of Linguistics • Burnaby, BC, Canada
- Ran **GLMMs and power analyses** for speech-perception studies; co-author on five papers.

Summer 2018 — **Data Scientist (Co-op)**
Statistics Canada • Ottawa, ON, Canada
- Evaluated a machine-learning record-linkage method by **Monte Carlo simulation**.

Education

2023 – 2027 — **Ph.D., Statistics and Data Science (expected summer 2027)**
University of Texas at Arlington • Arlington, TX, USA
- Dissertation: **Bayesian** modeling for partially observed and aggregate data.
- Teaching Assistant: Bayesian Data Analysis (graduate).

2017 – 2019 — **M.Sc., Statistics**
Simon Fraser University • Burnaby, BC, Canada
- Thesis: Covariance-adjusted sparse reduced-rank regression for imaging-genetics data (ADNI).

2013 – 2017 — **B.Sc., Applied Statistics, with Distinction**
University of Toronto • Toronto, ON, Canada

Skills

Methods — Bayesian hierarchical models, MCMC, Bayesian borrowing; indirect treatment comparisons (MAIC, STC); external control arms; survival analysis; GLMMs, GAMMs; simulation-based power analysis

Software — R (8+ years; Shiny, R Markdown), SAS, Python, SQL, LaTeX, Git, HPC (TACC, Compute Canada)

Publications

Methods (First Author)

1. **Ruan, H.**, Wang, K., Lim, J., & Wang, X. (2026). Bias meets Bayes: Bayesian aggregation for peer and self ranking [Accepted]. *Journal of the Royal Statistical Society Series A*.

Drug Development and Comparative Effectiveness

1. De Santis, M., Breijo, S. M., Robinson, P., Capone, C., Pascoe, K., Van Sanden, S., Hashim, M., Trevisan, M., Daly, C., Reitsma, F., van Beekhuizen, S., **Ruan, H.**, Heeg, B., & Verzoni, E. (2024). Feasibility of indirect treatment comparisons between niraparib plus abiraterone acetate and other first-line poly ADP-ribose polymerase inhibitor treatment regimens for patients with BRCA1/2 mutation-positive metastatic castration-resistant prostate cancer. *Advances in Therapy*, *41*(8), 3039–3058.
2. Strüebing, A., McKibbon, C., **Ruan, H.**, Mackay, E., Dennis, N., Velummailum, R., He, P., Tanaka, Y., Xiong, Y., Springford, A., & Rosenlund, M. (2024). Augmenting external control arms using Bayesian borrowing: A case study in first-line non-small cell lung cancer. *Journal of Comparative Effectiveness Research*, *13*(5), e230175.
3. He, J., Berringer, H., Heeg, B., **Ruan, H.**, Kampfenkel, T., Dwarakanathan, H., et al. (2022). Indirect treatment comparison of daratumumab, pomalidomide, and dexamethasone versus standard of care in patients with difficult-to-treat relapsed/refractory multiple myeloma. *Advances in Therapy*, *39*(9), 4230–4249.

Conference Abstracts

1. **Ruan, H.**, Springford, A., Gupta, A., & Mackay, E. (2022). MSR102 Variance-Bias Trade-Off in Covariate Adjustment in the Context of Synthetic Control Methods. *Value in Health*, *25*(12), S369.

In Preparation

1. **Ruan, H.**, et al. (in preparation). *Bayesian binomial decomposition for contaminated counts, with application to oral reading fluency assessment*.
2. **Ruan, H.**, et al. (in preparation). *A Bayesian predictive index for survival-data analysis readiness*.

Statistical Collaborations in Linguistics

Speech Communication (2025); *Int J Speech Technol* (2023, two papers); *Speech Prosody* (2026, two papers).

Presentation Awards and Service

2026, 2025 — **Third Place** (twice), ASA North Texas student presentations.

2019 — **Best Graduate Poster Presentation**, Canadian Statistics Student Conference.

2025 – Present — **President**, National Student Data Corps UTA Chapter; **Treasurer and Co-founder**, ASA Student Chapter at UTA.